

Lab Assignment-7.5

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Batch:03

Task 1 (Mutable Default Argument – Function Bug)

Task: Analyze given code where a mutable default argument causes unexpected behavior. Use AI to fix it.

Bug: Mutable default argument

```
def add_item(item, items=[]):
```

```
    items.append(item)
```

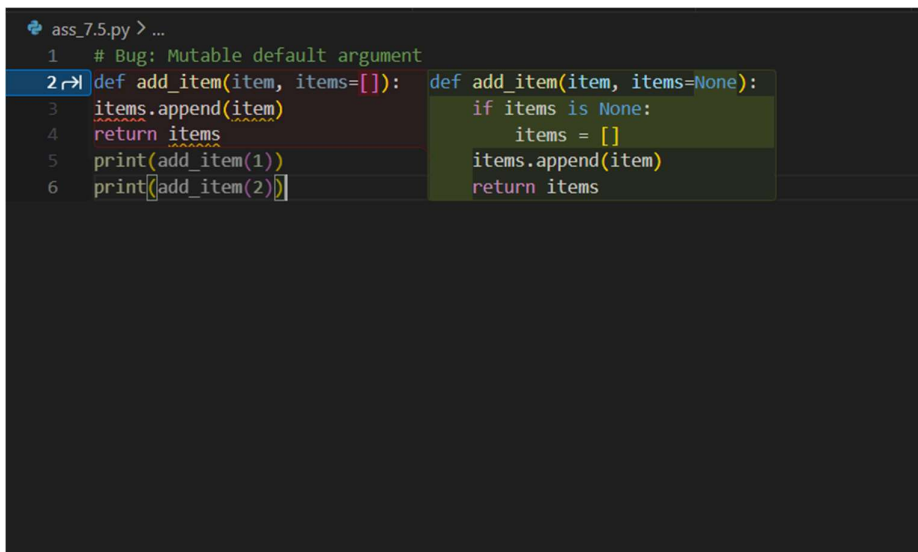
```
    return items
```

```
print(add_item(1))
```

```
print(add_item(2))
```

Expected Output: Corrected function avoids shared list bug.

Screenshots:



```
ass_7.5.py > ...
1 # Bug: Mutable default argument
2 def add_item(item, items=[]):
3     items.append(item)
4     return items
5 print(add_item(1))
6 print(add_item(2))

def add_item(item, items=None):
    if items is None:
        items = []
    items.append(item)
    return items
```

```

assg_07.py > ...
1  def add_item(item, items=None):
2      if items is None:
3          items = []
4      items.append(item)
5      return items
6  print(add_item(1))
7  print(add_item(2))

```

output:

```

PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS  GITLENS
Python Debug Console + - [ ] [ ] ...
PS C:\Users\91938\OneDrive\Documents\Desktop\AI> & 'c:\Users\91938\AppData\Local\Programs\Python\Python313\python.exe' 'c:\Users\91938\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-x64\bundle\libs\debugpy\launcher' '54845' '--' 'c:\Users\91938\OneDrive\Documents\Desktop\AI\AIAC7_5.py'
[1]
[2]

```

Task 2 (Floating-Point Precision Error)

Task: Analyze given code where floating-point comparison fails.

Use AI to correct with tolerance.

Bug: Floating point precision issue

```

def check_sum():
    return (0.1 + 0.2) == 0.3

```

```
print(check_sum())
```

Expected Output: Corrected function

Screenshots:

```

8
Fix the attached problems floating point prcision issue
Expected indented block Code is structurally... Auto
Add to chat (Alt+X) | Do not show again
9 def check_sum():
10     return (0.1 + 0.2) == 0.3
11     print(check_sum())

```

```
import math

def check_sum():
    return math.isclose(0.1 + 0.2, 0.3)

print(check_sum())
```

output:

```
PS C:\Users\91938\OneDrive\Documents\Desktop\AI> c::; cd 'c:\Users\91938\OneDrive\Documents\Desktop\AI'; & 'c:\Users\91938\AppData\Local\Programs\Python\Python313\python.exe' 'c:\Users\91938\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-x64\bundled\libs\debugpy\launcher' '60525' '--' 'C:\Users\91938\OneDrive\Documents\Desktop\AI\AIAC7_5.py'
True
PS C:\Users\91938\OneDrive\Documents\Desktop\AI>
```

Task 3 (Recursion Error – Missing Base Case)

Task: Analyze given code where recursion runs infinitely due to missing base case. Use AI to fix.

Bug: No base case

```
def countdown(n):
```

```
    print(n)
```

```
    return countdown(n-1)
```

```
countdown(5)
```

Expected Output : Correct recursion with stopping condition.

Screenshots:

```
ass_7.5.py > ...
3  print(n)
   ↗ if n == 0:
     return
4  return countdown(n-1)
5  countdown(5)
```

```

14
15 def countdown(n):
16     if n <= 0:
17         return
18     print(n)
19     return countdown(n-1)
20 countdown(5)

```

output:

```

PS C:\Users\91938\OneDrive\Documents\Desktop\AI> & 'c:\Users\91938\AppData\Local\Programs\Python\Python313\python.exe' 'c:\Users\91938\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-x64\bundle\libs\debugpy\launcher' '57677' '--' 'c:\Users\91938\OneDrive\Documents\Desktop\AI\AIAC7_5.py'
5
4
3
2
1
PS C:\Users\91938\OneDrive\Documents\Desktop\AI>

```

Task 4 (Dictionary Key Error)

Task: Analyze given code where a missing dictionary key causes error. Use AI to fix it.

Bug: Accessing non-existing key

```

def get_value():
    data = {"a": 1, "b": 2}
    return data["c"]
print(get_value())

```

Expected Output: Corrected with .get() or error handling.

Screenshots:

```

22 def get_value():
23     data = {"a": 1, "b": 2}
24     return data.get("c", "Key not found")
25 print(get_value())

```

```

21
22 def get_value():
23     data = {"a": 1, "b": 2}
24     return data.get("c", "Key not found")
25 print(get_value())

```

output:

```

1
PS C:\Users\91938\OneDrive\Documents\Desktop\AI> c:: cd 'c:\Users\91938\OneDrive\Documents\Desktop\AI'; & 'c:\Users\91938\AppData\Local\Programs\Python\Python313\python.exe' 'c:\Users\91938\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-x64\bundle\libs\debugpy\launcher' '52379' '--' 'c:\Users\91938\OneDrive\Documents\Desktop\AI\AIAC7_5.py'
Key not found
PS C:\Users\91938\OneDrive\Documents\Desktop\AI>

```

Task 5 (Infinite Loop – Wrong Condition)

Task: Analyze given code where loop never ends. Use AI to detect

and fix it.

Bug: Infinite loop

```
def loop_example():
```

```
    i = 0
```

```
    while i < 5:
```

```
        print(i)
```

Expected Output: Corrected loop increments i.

Screenshots:

```

27
28 def loop_example():
29     i = 0
30     while i < 5:
31         print(i)
        i += 1

```

```

28
29 def loop_example():
30     i = 0
31     while i < 5:
32         print(i)
33         i += 1
34 loop_example()

```

output:

```

PS C:\Users\91938\OneDrive\Documents\Desktop\AI> c:: cd 'c:\Users\91938\OneDrive\Documents\Desktop\AI'; & 'c:\Users\91938\AppData\Local\Programs\Python\Python313\python.exe' 'c:\Users\91938\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-x64\bundle\libs\debugpy\launcher' '58963' '--' 'c:\Users\91938\OneDrive\Documents\Desktop\AI\AIAC7_5.py'
0
1
2
3
4

```

Task 6 (Unpacking Error – Wrong Variables)

Task: Analyze given code where tuple unpacking fails. Use AI to fix it.

Bug: Wrong unpacking

a, b = (1, 2, 3)

Expected Output: Correct unpacking or using _ for extra values.

Screenshots:

```

36
37 a, b, _ = (1, 2, 3)
38 print(a, b)

```

output:

```

PS C:\Users\91938\OneDrive\Documents\Desktop\AI> c:: cd 'c:\Users\91938\OneDrive\Documents\Desktop\AI'; & 'c:\Users\91938\AppData\Local\Programs\Python\Python313\python.exe' 'c:\Users\91938\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-x64\bundle\libs\debugpy\launcher' '55028' '--' 'c:\Users\91938\OneDrive\Documents\Desktop\AI\AIAC7_5.py'
1 2
PS C:\Users\91938\OneDrive\Documents\Desktop\AI>

```

Task 7 (Mixed Indentation – Tabs vs Spaces)

Task: Analyze given code where mixed indentation breaks execution. Use AI to fix it.

Bug: Mixed indentation

```
def func():
```

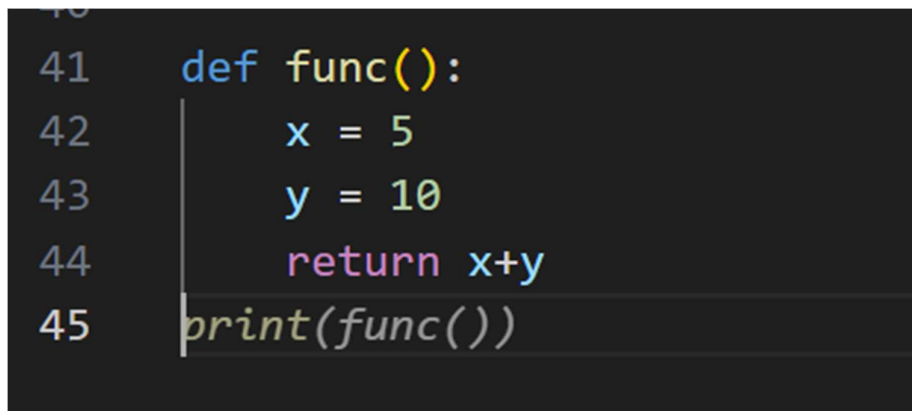
```
    x = 5
```

```
    y = 10
```

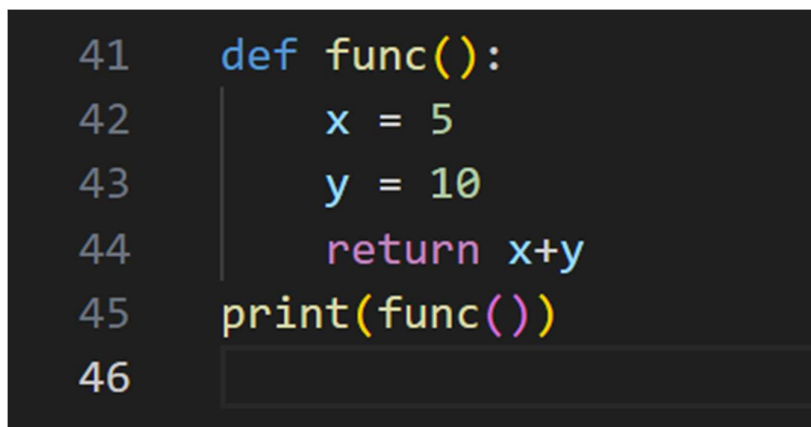
```
    return x+y
```

Expected Output : Consistent indentation applied.

Screenshots:

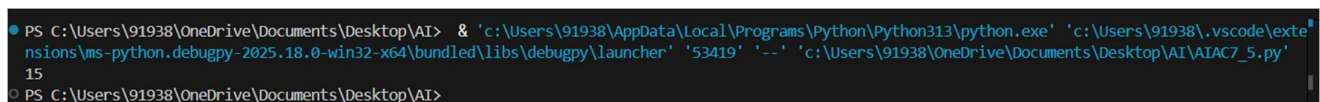


```
41  def func():
42      x = 5
43      y = 10
44      return x+y
45  print(func())
```



```
41  def func():
42      x = 5
43      y = 10
44      return x+y
45  print(func())
46
```

output:



```
PS C:\Users\91938\OneDrive\Documents\Desktop\AI> & 'c:\Users\91938\AppData\Local\Programs\Python\Python313\python.exe' 'c:\Users\91938\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-x64\bundled\libs\debugpy\launcher' '53419' '--' 'c:\Users\91938\OneDrive\Documents\Desktop\AI\AIAC7_5.py'
15
PS C:\Users\91938\OneDrive\Documents\Desktop\AI>
```

Task 8 (Import Error – Wrong Module Usage)

Task: Analyze given code with incorrect import. Use AI to fix.

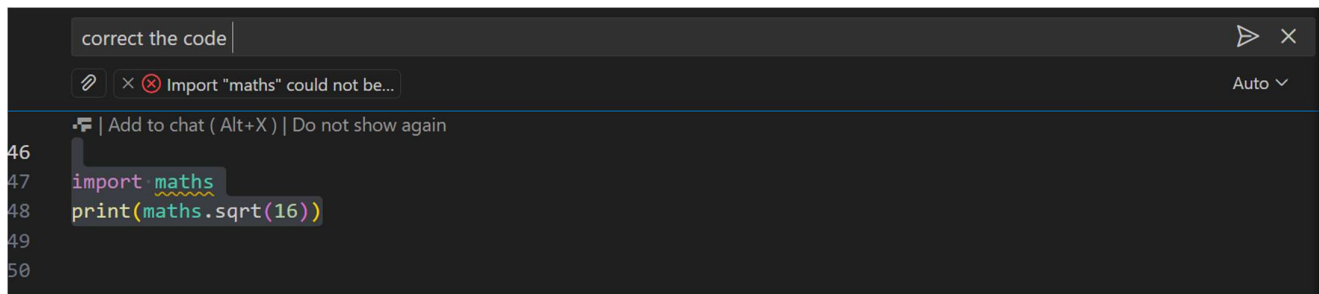
Bug: Wrong import

```
import maths
```

```
print(maths.sqrt(16))
```

Expected Output: Corrected to import math

Screenshots:

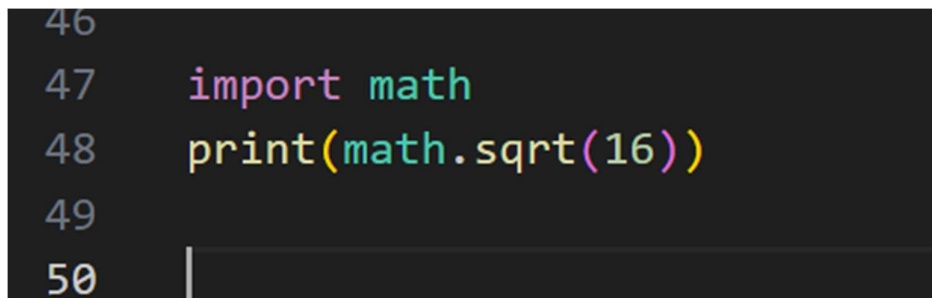


The screenshot shows a VS Code editor window with a dark theme. At the top, there is a search bar with the text "correct the code". Below it, a notification bar displays a red 'x' icon and the message "Import 'maths' could not be...". The main editor area shows a Python script with the following code:

```
46  
47 import maths  
48 print(maths.sqrt(16))  
49  
50
```

The code is highlighted with a blue selection box. A tooltip or suggestion box is visible over the code, showing the corrected code:

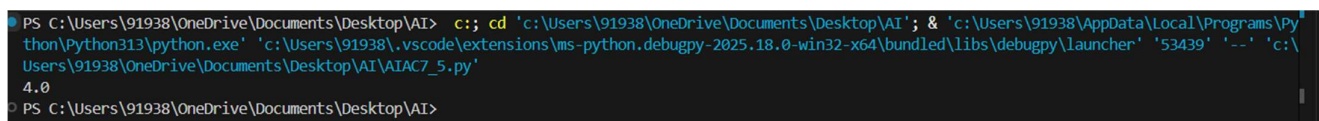
```
import math  
print(math.sqrt(16))
```



The screenshot shows the corrected Python code in a dark-themed editor:

```
46  
47 import math  
48 print(math.sqrt(16))  
49  
50
```

output:



The screenshot shows a terminal window with the following command and output:

```
PS C:\Users\91938\OneDrive\Documents\Desktop\AI> c:: cd 'c:\Users\91938\OneDrive\Documents\Desktop\AI'; & 'c:\Users\91938\AppData\Local\Programs\Python\Python313\python.exe' 'c:\Users\91938\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-x64\bundled\libs\debugpy\launcher' '53439' '--' 'c:\Users\91938\OneDrive\Documents\Desktop\AI\AIAC7_5.py'  
4.0  
PS C:\Users\91938\OneDrive\Documents\Desktop\AI>
```